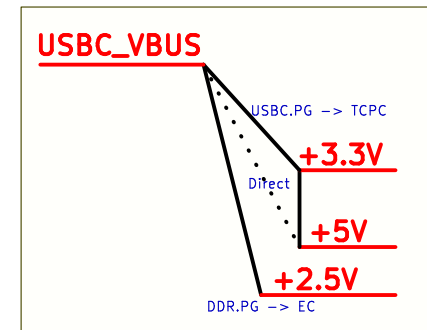
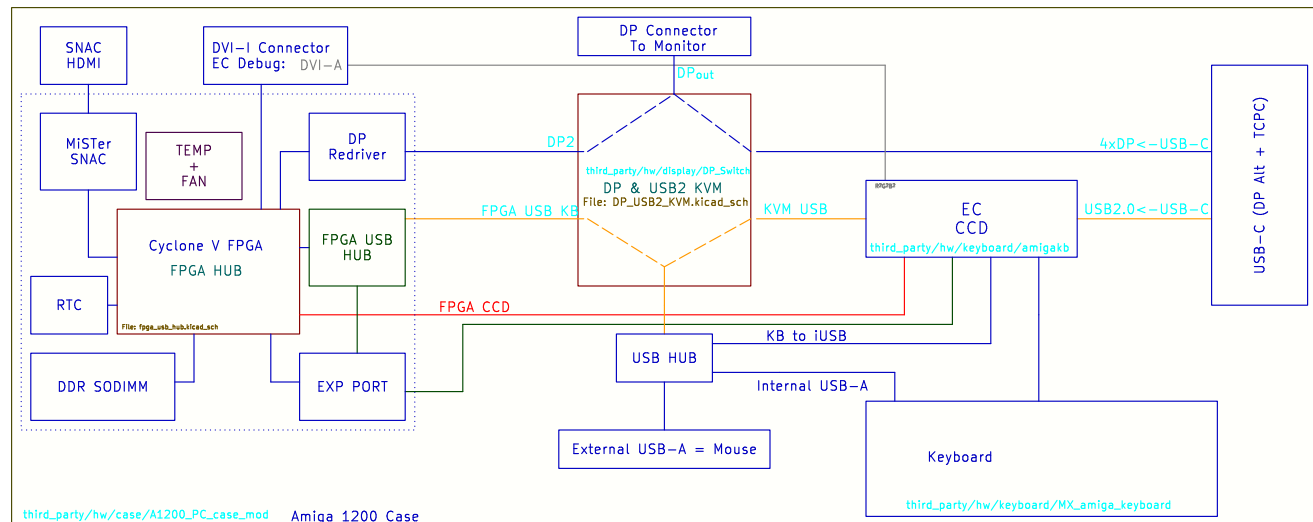




FPGA function and pin assignments

* DP on HDMI Flexlt
Net : FPGA Pin
DP0- : AD11
DP0+ : AD10
DP1- : AH5
DP1+ : AH6
AUX- : AE7
AUX+ : AF6
HPD : AF11

* SNAC on HDMI
Pin: FPGA Pin
1 : AH9
2 : AG11
3 : AF15
4 : AH11
5 : AH12
6 : AG16
7 : AF17
8 : AD4

* Input Switches
Net : FPGA Pin
OSD : AF13
RST : AG13
USER: U13

* LED outputs
Net : FPGA Pin
PWR : AA23
USR : AE26
DSK : Y16

* DP on GPIO_1 header
Net : FPGA Pin
DP0- : AA15
DP0+ : Y15
DP1- : AF28
DP1+ : AF27
DP2- : AH27
DP2+ : AG28
AUX- : AD26
AUX+ : AE25
HPD : AC24
LP : AG26

* I2S for Line Out
Net : FPGA Pin
DIN : AC4
LRCK: AG8
BCK : AG10
MUTE: AH8

* SLiRP UART
Net : FPGA Pin
RXD : U9
TXD : V10

* I2C Controller
Net : FPGA Pin
SDA : AG9
SCK : U14

* DVI-D
Net : FPGA Pin
D0+ : AG25
D0- : AF25
D1+ : AH24
D1- : AG24
D2+ : AG23
D2- : AF23
CK+ : AH23
CK- : AH22
HPD : AG21

ESP32/EC Functions and pin assignments

* Boot strap (flash): IO0@i, EN@i, U0RXD@i, U0TXD@o
* TCPC
* Negotiate SINK 5V@3A + 4-lane DP Alt Mode
* I2C Controller: IO21/SDA, IO38/SCL
* DVI-A/VGA output:
* Red[0,1] = [IO10,IO11]@o3v3
* Green[0,1] = [IO45,IO40]@o3v3
* Blue[0,1] = [IO46,IO39]@o3v3
* HSYNC = IO47@o1v8
* VSYNC = IO48@o1v8
* DDR
* Power Good: IO2@o3v3
* I2C same pin as controller
* WiFi SLiRP modem, UART1
* TxD: U1TXD@o3v3
* RxD: U1RXD@i3v3
* KVM
* User Switch: IO3@i falling edge
* Port selection: IO5@o3v3, L=USBC DP, H=FPGA DP
* FPGA
* Temp sensor interrupt alarm IO4@i falling edge
* Cooling Fan: SENS=IO7@i, PWM=IO8@o3v3
* I2C peripheral: IO41/SCL, IO42/SDA
Expansion port:
* ADC1_CH8
A500 Keyboard:
* CLK=IO12@o3v3
* DATA=IO13@i3v3
* RST=IO14@o3v3
* LED STATUS=IO15@o3v3
* LED INUSE=IO16@o3v3

B1200

CC BY-NC-SA 4.0
<https://creativecommons.org/licenses/by-nc-sa/4.0/>
DRAFT



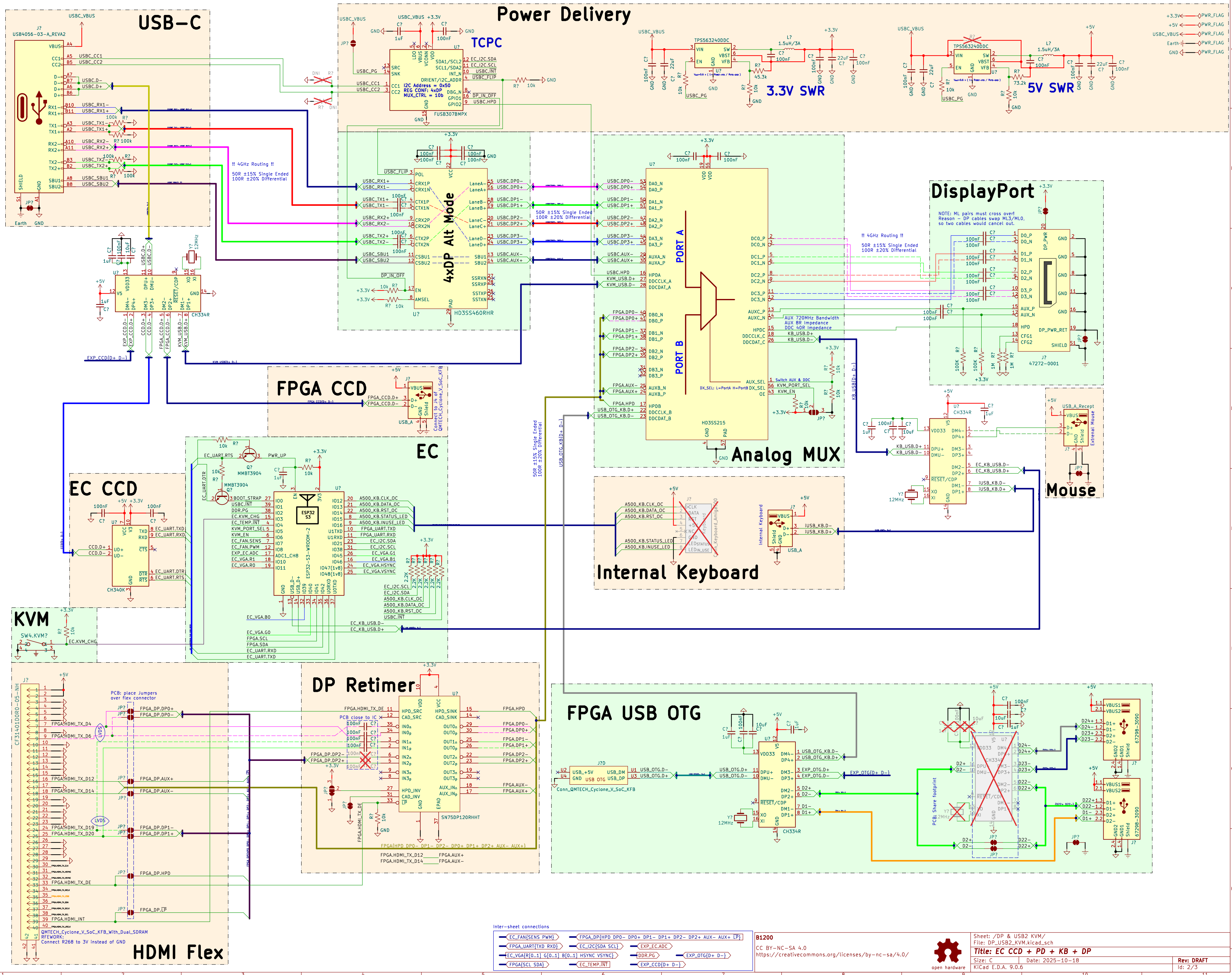
open hardware

Sheet: /
File: b1200.kicad_sch

Title: Block Diagram

Size: A Date: 2025-10-18
KiCad E.D.A. 9.0.6

Rev: DRAFT
Id: 1/3



Inter-sheet connections

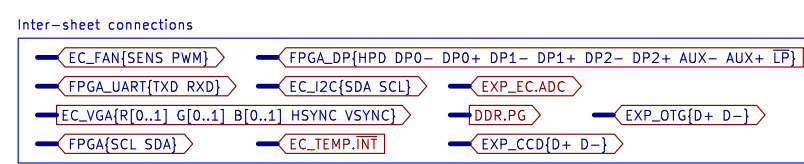
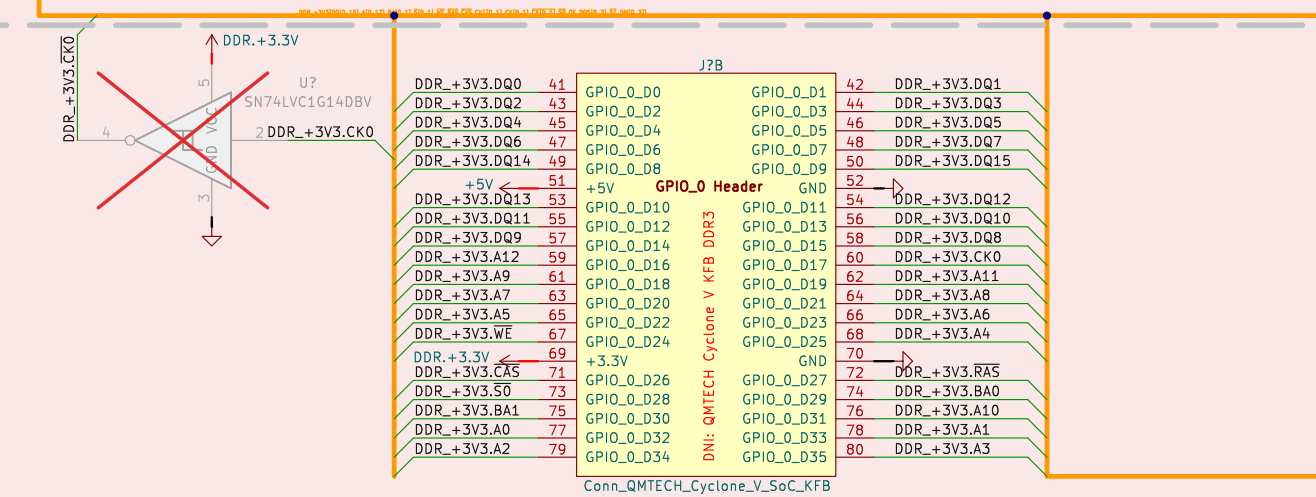
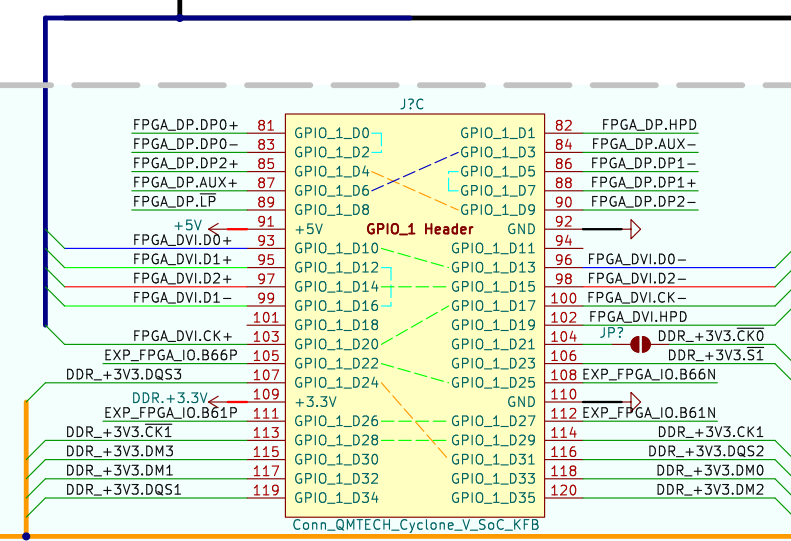
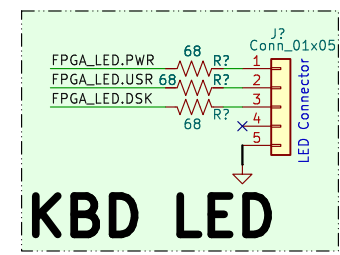
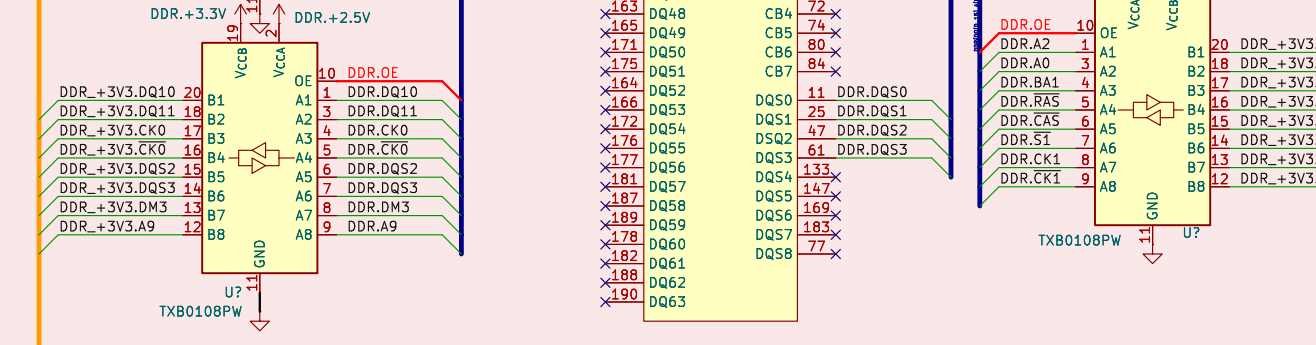
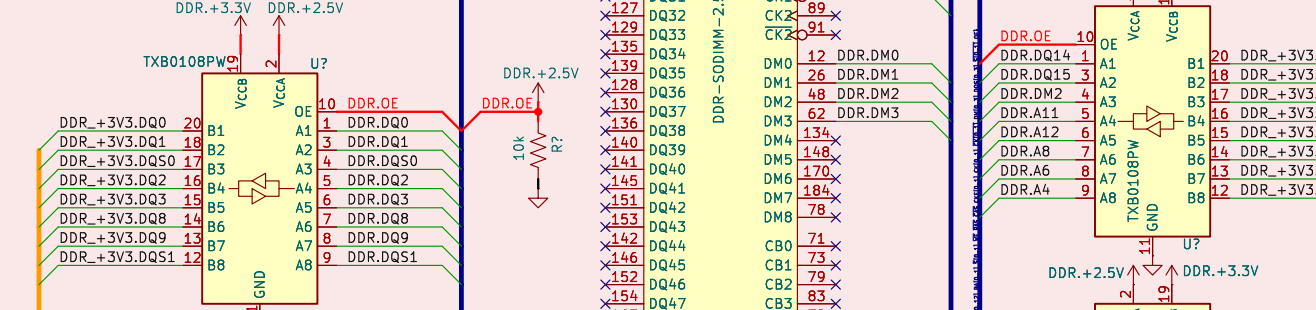
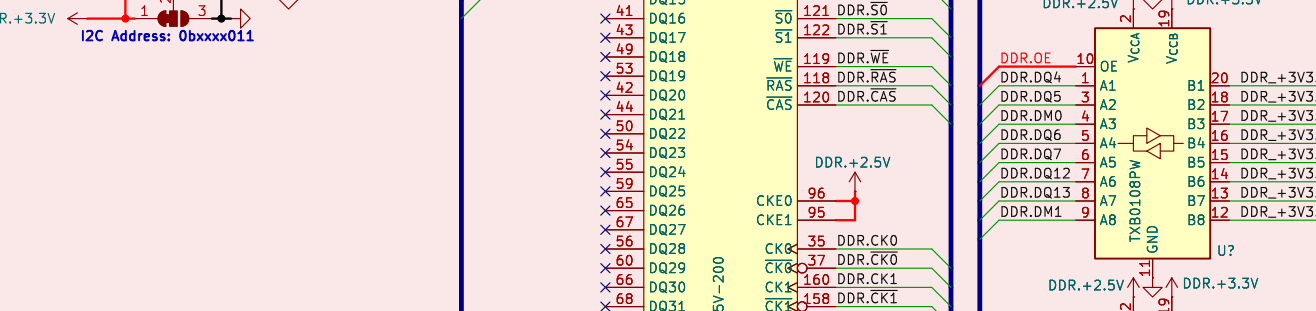
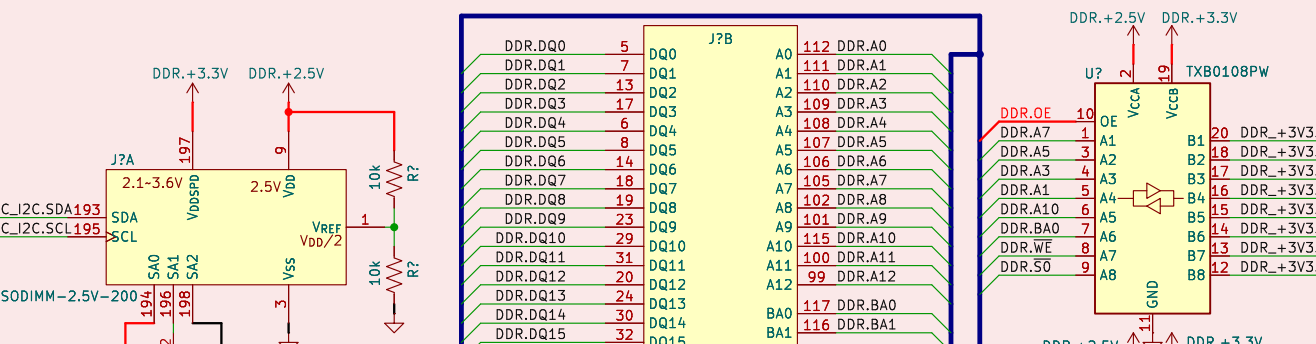
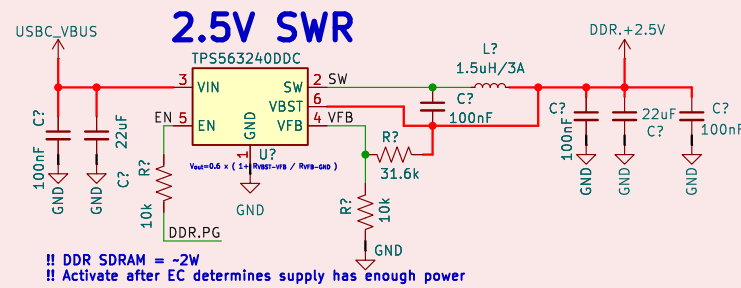
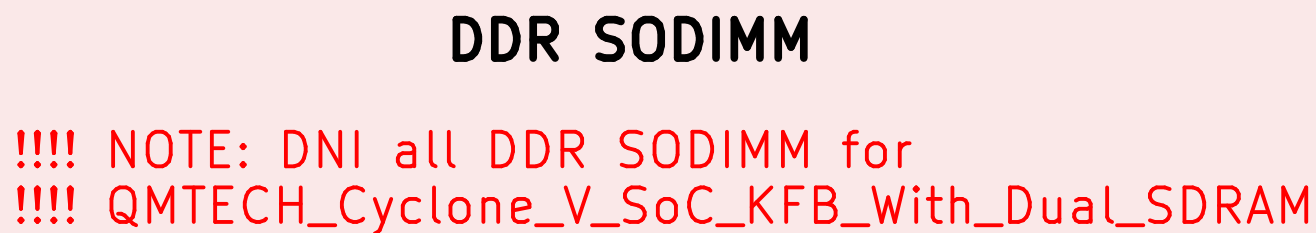
- EC_FAN[SENS PWM] → FPGA_DP[HPD DP0- DP0+ DP1- DP1+ DP2- DP2+ AUX- AUX+ LP]
- FPGA_UART[TXD RXD] → EC_I2C[SCL SDA]
- EC_VGA[R[0..1] G[0..1] B[0..1] HSYNC VSYNC] → DDR_PG
- FPGA[SCL SDA] → EC_TEMP[INT]
- FPGA_DP[HPD DP0- DP0+ DP1- DP1+ DP2- DP2+ AUX- AUX+ LP] → EXP_OTG[D+ D-]
- EC_I2C[SCL SDA] → EXP_OTG[D+ D-]
- DDR_PG → EXP_OTG[D+ D-]
- EXP_OTG[D+ D-] → EXP_OTG[D+ D-]

B1200

CC BY-NC-SA 4.0
<https://creativecommons.org/licenses/by-nc-sa/4.0/>



Sheet: /DP & USB2 KVM/
File: DP_USB2_KVM.kicad_sch
Title: EC CCD + PD + KB + DP
Size: C Date: 2025-10-18
KiCad E.D.A. 9.0.6
Rev: DRAFT
Id: 2/3



B1200
CC BY-NC-SA 4.0
<https://creativecommons.org/licenses/by-nc-sa/4.0/>



Sheet: /FPGA HUB/
File: fpga_usb_hub.kicad_sch

Title: FPGA + DRAM + Audio + RTC + FAN + SNAC

Size: C	Date: 2025-10-18
---------	------------------

KiCad E.D.A. 9.0.6

Rev: DRAFT
Id: 3/3